Per-Example Predictive Multiplicity in Hybrid Linear Recurrent Architectures

Anonymous ACL submission

Abstract

Practitioners select among hybrid linear recurrent architectures using aggregate benchmarks, but these scores can mask per-example consequences. Across three controlled designs, adding attention affects substantially different examples per architecture (Jaccard ≈ 0.10, approximately twice the independence baseline); validation across five architectures and ten pairwise comparisons confirms this on all four benchmarks (all $p < .01$). Using the model zoo of Wang et al. (2025) (three architectures, five attention ratios, two scales), we decompose per-example effects and find that this predictive multiplicity (Marx et al., 2020) depends on task type ($\alpha = 0.14$–0.30) rather than attention dose. Aggregate trends do not reach statistical significance ($n = 5$; Bonferroni-adjusted $p = .170$), yet per-example overlap saturates at 4% attention and remains flat across all four benchmarks. Majority vote fails to exploit this complementarity, confirming that configurations interchangeable by aggregate metrics carry hidden per-example consequences. Confidence-informed routing extracts 36% of oracle complementarity on LAMBADA (+5.3 pp), suggesting these differences are partially actionable.

1 Introduction

Hybrid recurrent architectures that appear equivalent by aggregate benchmarks affect largely non-overlapping sets of individual examples: adding attention to three controlled linear recurrent backbones yields helped and hurt sets with at most 10% pairwise overlap, approximately twice the independence baseline. This per-example divergence is invisible in aggregate metrics: on LAMBADA, cross-architecture standard deviation drops 79% as attention increases (Bonferroni-adjusted $p = .170$, BCa CI $[-1.0, -0.7]$), but each architecture reaches similar accuracy by gaining and losing accuracy on different inputs. Practitioners increasingly choose between recurrent backbones based on aggregate benchmark scores. Across all ten pairwise comparisons among five architectures, overlap in per-example effects remains below 14% (all $p < .01$), meaning the choice of backbone is associated with which specific inputs a deployed system handles correctly.

This tension arises in the context of a rapid architectural shift toward hybrid recurrent-attention models, which interleave attention layers within linear recurrent backbones. Jamba (Lieber et al., 2024), Griffin (De et al., 2024), RWKV-6 (Peng et al., 2024), and RecurrentGemma (Botev et al., 2024) each combine recurrence with attention using different mixing strategies. These systems now serve as production-grade alternatives to pure Transformers, and the practical question of how much attention to include is typically answered by comparing aggregate benchmark numbers. Wang et al. (2025) provide a controlled model zoo spanning three recurrent architectures, five attention ratios, and two parameter scales, which enables systematic study of how attention allocation affects these hybrid designs.

Most evaluations of hybrid architectures compare aggregate accuracy, which can mask per-example behavioral diversity. Prior work on predictive multiplicity (Marx et al., 2020) and underspecification (D'Amour et al., 2022; McCoy et al., 2020) documents this risk for seed variation within a single architecture, but whether the phenomenon extends to architectural variation in hybrid recurrent models remains open. If architectures that score identically in aggregate behave differently on individual inputs, the choice of backbone carries hidden consequences for deployment reliability and ensemble construction.

We present the first systematic per-example analysis of how attention affects cross-architecture agreement in hybrid recurrent models. Using the model zoo of Wang et al. (2025), we compare three

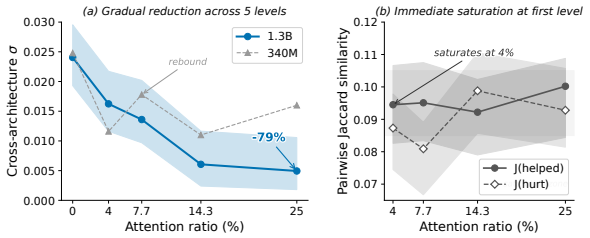


Figure 1: Aggregate convergence and per-example divergence are decoupled (helped: correct only with attention; hurt: correct only without). (a) Cross-architecture standard deviation on LAMBADA decreases gradually with attention ratio at 1.3B (solid) but partially rebounds at 340M (dashed). (b) Cross-architecture pairwise Jaccard similarity of per-example helped/hurt sets saturates at the first attention level (4%) and remains flat. Shaded bands are illustrative ($n = 3$).

architectures (GatedDeltaNet, RetNet, HGRN2) at five attention ratios (0% to 25%) and two scales (340M and 1.3B parameters). These three architectures represent distinct approaches to linear recurrence: gated delta rules, multi-scale exponential decay, and gated state expansion. We evaluate all 30 configurations on 10 benchmarks spanning retrieval, comprehension, reasoning, and mixed task categories. On four benchmarks (LAMBADA, $N = 5{,}153$; HellaSwag, $N = 10{,}042$; PIQA, $N = 1{,}838$; ARC-Easy, $N = 2{,}376$) we conduct per-example decompositions, classifying each example as helped, hurt, or unchanged when attention is added, and measure cross-architecture overlap using Jaccard similarity. Figure 1 previews our central result: aggregate convergence and per-example divergence follow different trajectories as attention ratio increases.

Our analysis yields two findings. First, per-example predictive multiplicity varies in magnitude across tasks, forming a task-dependent variation. The ambiguity α (Marx et al., 2020) ranges from 0.14 (PIQA) through 0.15 (HellaSwag) and 0.21 (ARC-Easy) to 0.30 (LAMBADA), while pairwise Jaccard remains uniformly low (≤ 0.10, all $p < .01$). Within each benchmark, α is stable across attention ratios (variation < 0.03), but differs substantially across task types, ranging from 0.14 (reasoning) to 0.30 (retrieval).

Aggregate convergence is task-specific and exploratory: of 10 benchmarks, only LAMBADA shows monotonic variance reduction ($\rho = -1.0$, Bonferroni-adjusted $p = .170$; observed at 1.3B, absent at 340M). Yet on this benchmark, per-example Jaccard saturates at the first attention level while aggregate variance continues to decrease (§4.2).

Our contributions are as follows:

- We show that architectural variation under matched training conditions is associated with structured per-example predictive multiplicity ($\alpha = 0.14$–0.30) across four benchmarks, with magnitude determined by task type rather than attention dose (§4.1, Table 2). Extended validation across five architectures confirms the pattern: all ten pairwise Jaccard values remain ≤ 0.14 across four benchmarks (all $p < .01$; Appendix X). The primary controlled variable is the recurrent backbone (cf. McCoy et al., 2020; D'Amour et al., 2022), though single training runs prevent full disentanglement from seed effects (Limitations).

- Per-example Jaccard saturates at the first attention level (4%) across all four benchmarks and shows no further increase at higher ratios, while aggregate variance shows a non-significant decreasing pattern on LAMBADA (§4.2; Appendix N). This immediate saturation indicates that minimal attention suffices to establish the full pattern of configuration-specific per-example effects; additional attention narrows aggregate gaps without homogenizing which examples each architecture affects. The aggregate convergence component is dependent on GDN inclusion (§4.3) and does not replicate on other benchmarks.

2 Related Work

Hybrid Recurrent Architectures. The design space for combining attention with recurrence spans layer-level interleaving (Jamba; Lieber et al., 2024), local attention pairing (Griffin, RecurrentGemma; De et al., 2024; Botev et al., 2024), matrix-valued recurrence (RWKV-6; Peng et al., 2024), within-head mixing (Hymba; Dong et al., 2024), and unbounded-context hybrid SSMs (Samba; Ren et al., 2024). H3 (Fu et al., 2023) and Hyena (Poli et al., 2023) establish early design points in the hybrid space with gated convolutions and subquadratic operators. Mamba-2 (Dao and Gu, 2024) establishes a duality between structured state spaces and attention; Mamba-3 (Lahoti et al., 2026) extends the design further, while Ali et al. (2025) reveal implicit attention within Mamba. Comparative evaluations confirm that hybrids outperform pure

Transformers and SSMs at scale (Waleffe et al., 2024), and that even one attention layer enables content-based routing (Basu, 2026). Wang et al. (2025) provide the controlled model zoo we use, holding training conditions constant while varying architecture and attention ratio.

Linear Recurrent Models. Our three recurrent backbones represent distinct gating strategies within the linear recurrence family: RetNet (Sun et al., 2023) uses multi-scale exponential decay, HGRN2 (Qin et al., 2024) applies gated state expansion, and GatedDeltaNet (Yang et al., 2024) combines gated updates with delta rules. These descend from S4 (Gu et al., 2022) and Mamba (Gu and Dao, 2024), which introduced structured state spaces and input-dependent selection, respectively, and connect to linear attention (Katharopoulos et al., 2020) through the state-space–attention duality of Mamba-2.

Underspecification and Predictive Multiplicity. D'Amour et al. (2022) formalized underspecification: equivalent aggregate performance can mask per-prediction divergence. Concrete NLP instances include seed-dependent BERT disagreement (McCoy et al., 2020; Sellam et al., 2022; Dodge et al., 2020) and the four-phase convergence pattern of autoregressive LMs (Fehlauer et al., 2025). Marx et al. (2020) introduced predictive multiplicity as a complementary lens (extending prediction churn; Milani Fard et al. 2016), showing that near-optimal models produce conflicting per-example predictions; extensions address probabilistic classification (Watson-Daniels et al., 2023), Rashomon capacity (Hsu and Calmon, 2022), reconciliation (Du et al., 2025; Haghighat et al., 2026), and cross-model disagreement as a correctness signal (Gorbett and Jana, 2026). Swayamdipta et al. (2020) characterize examples via training dynamics (variability across epochs); our decomposition operates at deployment time across architectures with no training signal. Niven and Kao (2019) show that benchmark artifacts can inflate per-example agreement; our cross-benchmark variation ($\alpha = 0.14$–0.30) suggests the patterns we observe are not driven by a single artifact type. Fort et al. (2019) provide a loss landscape perspective on deep ensembles, showing that diverse architectures explore distinct modes and produce disagreement on individual predictions; standard ensemble diversity measures (Kuncheva and Whitaker, 2003), including the Q-statistic (Brown et al., 2005), quantify such disagreement; Jaccard similarity is one instance. Black et al. (2022) survey the model multiplicity landscape and Cooper et al. (2024) connect predictive arbitrariness to fairness. These studies attribute divergence to initialization noise or the near-optimal set within a single architecture. Whereas Rashomon-set analyses (Fisher et al., 2019; Semenova et al., 2022; Jain et al., 2025) examine variation among near-optimal models within a single hypothesis class, we study variation across architectural families under matched training conditions. We extend this line of work to a new source of variation, the recurrent backbone itself, showing that architectural variation is associated with per-example multiplicity that varies systematically by task type. Unlike ensemble diversity, which is designed into the system, the per-example divergence we observe emerges from architectural choice without explicit diversification. The observation that aggregate-level associations can diverge from individual-level patterns is a well-known statistical phenomenon (ecological fallacy; Robinson 1950). Our decoupling finding (§4.2) is a specific instance: aggregate variance decreases while per-example overlap remains flat. The contribution lies not in the conceptual observation but in its quantitative characterization across architectures and task types.

3 Method

We describe the model configurations, evaluation benchmarks, and analytical framework used to measure how attention affects cross-architecture agreement at both aggregate and per-example levels.

3.1 Model Zoo

We use the pretrained model zoo of Wang et al. (2025), which provides controlled comparisons across architectures and attention ratios. The zoo contains three linear recurrent architectures: GatedDeltaNet (Yang et al., 2024), RetNet (Sun et al., 2023), and HGRN2 (Qin et al., 2024). These three architectures represent distinct approaches to linear recurrence: GatedDeltaNet uses gated delta rules, RetNet uses multi-scale exponential decay, and HGRN2 uses gated linear recurrence with state expansion. Each architecture is trained at five attention ratios: 0% (pure recurrent), 4%, 7.7%, 14.3%, and 25%. We adopt the hk-1 notation from the original work, where one attention layer is interleaved per every k+1 total layers: h3-1 corresponds to

25% attention, h6-1 to 14.3%, h12-1 to 7.7%, and h24-1 to 4%. All configurations are available at two scales: 340M and 1.3B parameters. Within each scale, all models share identical training data, tokenizer, and hyperparameters; they differ only in architecture and attention configuration. This controlled design isolates the effect of attention from confounds such as training data or optimization differences. The zoo also includes a Transformer baseline (pure attention) at both scales.

3.2 Task Taxonomy

We evaluate on 10 benchmarks grouped into four categories defined by the primary cognitive demand of each task. This taxonomy was fixed prior to analysis following established distinctions in the hybrid architecture literature (Arora et al., 2024a) (see Appendix A for the confirmatory/exploratory split), not derived post hoc from observed results. The categories test whether the effect of attention on cross-architecture variance depends on task type.

- **Retrieval**: LAMBADA (Paperno et al., 2016), which requires predicting a final word from long-range context without reasoning or world knowledge.
- **Comprehension**: BoolQ, a reading comprehension task with yes/no answers.
- **Reasoning**: HellaSwag, ARC-Challenge (Clark et al., 2018), WinoGrande, and PIQA. COPA ($n = 100$) is supplementary; its small sample renders α unreliable (Appendix U).
- **Mixed**: ARC-Easy, MMLU, SciQ, and XStoryCloze, which combine retrieval, comprehension, and reasoning demands.

All evaluations use lm-evaluation-harness v0.4.x (Gao et al., 2024) with default prompts and greedy decoding; accuracy serves as the primary metric for all benchmarks. Bootstrap analyses use random seed 42 throughout.

3.3 Convergence Metrics

We quantify cross-architecture agreement as a function of attention ratio. For each benchmark and attention ratio, we compute the cross-architecture standard deviation σ over the three architectures' accuracy scores. To measure the relationship between attention and convergence, we compute the Spearman rank correlation ρ between attention ratio rank and σ across the five attention configurations ($n = 5$). A negative ρ indicates that higher attention ratios are associated with lower cross-architecture variance.

We also report the convergence ratio $\sigma_{\text{pure}}/\sigma_{\text{h3-1}}$, defined as the standard deviation under the pure-recurrent configuration divided by that under the highest attention ratio (25%). Values greater than 1 indicate that higher attention ratios are associated with lower cross-architecture variance.

3.4 Per-Example Decomposition

To move beyond aggregate metrics, we decompose performance changes at the individual-example level on four benchmarks spanning three task categories: LAMBADA (5,153 examples; retrieval), HellaSwag (10,042; reasoning), PIQA (1,838; reasoning, two-choice), and ARC-Easy (2,376; mixed). HellaSwag uses normalized accuracy (acc_norm) as the per-example correctness criterion. Supplementary analyses on BoolQ (3,270; comprehension) and WinoGrande (1,267; reasoning, two-choice) appear in Appendix R and S. For each architecture, we compare the pure-recurrent baseline against its h3-1 hybrid (25% attention), which maximizes the contrast between recurrent and hybrid modes; §4.1 confirms that α is stable across intermediate ratios (variation < 0.03). We classify each example into one of three categories:

- **Helped**: correct in the hybrid but incorrect in the pure model.
- **Hurt**: correct in the pure model but incorrect in the hybrid.
- **Unchanged**: both models agree (both correct or both incorrect).

The net effect for each architecture is $|\text{helped}| - |\text{hurt}|$. A positive net effect means more examples transition to correct than to incorrect when attention is added; a negative net effect means the opposite.

To measure cross-architecture overlap in per-example effects, we compute the pairwise Jaccard similarity $J = |A \cap B|/|A \cup B|$ between the helped sets of two architectures and, separately, between their hurt sets. $J = 1$ indicates identical sets; $J = 0$ indicates no overlap. Low J values would indicate that different architectures are affected on different subsets of examples, which would constitute per-example divergence even when aggregate

performance converges. On K-choice tasks, binarization collapses $K-1$ wrong answers into a single "incorrect" category, potentially counting architectures as agreeing (both wrong) when they select different wrong answers. This conservative bias strengthens the multiplicity claim: true per-example divergence may exceed what binary Jaccard captures. Appendix O confirms that results are robust under continuous Jaccard similarity, mitigating concerns about information loss from binarization.

3.5 Statistical Methods

All statistical tests are two-tailed unless otherwise noted.

Permutation test. With only $n = 5$ attention ratios, we use an exact permutation test for ρ. All $5! = 120$ permutations of the attention ratio ranks are enumerated, and the two-tailed p-value is the fraction of permutations yielding $|\rho| \geq |\rho_{\text{obs}}|$.

Power considerations. With $n = 5$ ordinal levels, only $\rho = \pm 1.0$ achieves $p < .05$ (exact $p = .017$); the test has no power to detect strong but imperfect monotonic trends ($|\rho| = 0.9$ yields $p = .083$). This binary resolution means that a single rank swap suffices to shift the verdict from significant to non-significant. Consequently, all inferences based on Spearman correlations in this study should be interpreted as descriptive and exploratory rather than confirmatory.

Bootstrap confidence intervals. We construct 95% BCa (bias-corrected and accelerated) bootstrap confidence intervals for ρ using 10,000 resamples at the architecture level. With only $n = 3$ architectures, the resampling space is limited (at most 27 unique resamples before considering ratio permutations), so these intervals should be interpreted as heuristic indicators of effect direction rather than intervals with formal coverage guarantees. When the BCa correction degenerates at a boundary value, we report percentile confidence intervals instead. These intervals provide a complementary characterization of effect magnitude alongside the permutation test.

Multiple comparisons. We apply Bonferroni correction across all 10 benchmarks (adjusted threshold $\alpha/10 = 0.005$). Six benchmarks were designated confirmatory and four exploratory prior to analysis (Appendix A); all share the 10-benchmark correction to ensure conservative inference.

Within- and cross-architecture tests. For within-architecture comparisons of per-example properties (e.g., token count of helped vs. hurt examples), we use the Mann-Whitney U test. For cross-architecture comparisons of per-example effect distributions, we use the Kruskal-Wallis test. Both are rank-based nonparametric tests, chosen because the per-example effect distributions are non-normal and contain ties.

Attention weight analysis. To probe whether attention mechanisms differ between helped and hurt examples, we extract attention weight matrices from each attention layer via forward hooks and compute two quantities: (1) cosine similarity between the mean attention distributions of two architectures at each layer, which measures cross-architecture agreement in attention allocation; and (2) Shannon entropy of the attention distribution for each example, which measures how concentrated or diffuse the attention pattern is. Details of the extraction procedure appear in Appendix F.

4 Experiments

Using the model zoo of Wang et al. (2025) (§3.1), we evaluate all 30 configurations on 10 benchmarks plus supplementary COPA ($n = 100$) at 1.3B scale unless noted. Results at 340M appear in scale analyses. Recall: hk-1 denotes one attention layer per $k+1$ layers; h3-1 = 25% attention.

4.1 Per-Example Divergence

Do architectures converge by improving on the same examples? We compare each architecture's pure-recurrent baseline against its h3-1 hybrid on 5,153 LAMBADA examples, classifying each as helped (correct only in hybrid), hurt (correct only in pure), or unchanged. Although the decomposition is binary, we verify below that it captures genuine separation in the continuous confidence space (Cohen's $d = 0.73$–0.92 between helped and hurt on LAMBADA; cross-benchmark details in Appendix P.1). This binarization discards confidence magnitude (a transition from 0.51 to 0.49 is treated identically to one from 0.99 to 0.01), so we supplement it with continuous confidence margin analysis (Appendix P) and verify that low Jaccard persists under continuous thresholds (Appendix O).

Each architecture displays a distinct profile of per-example changes (Table 9). GDN loses a

Table 1: Cross-architecture convergence across benchmarks (1.3B scale). ρ: Spearman correlation between attention ratio and cross-architecture σ across five configurations. p: exact permutation test (two-tailed, 120 permutations). σ values reported $\times 10^2$ (accuracy percentage points). Ratio: $\sigma_{\text{pure}}/\sigma_{\text{h3-1}}$. **Bold**: only benchmark with $\rho = -1.0$. HellaSwag evaluated using normalized accuracy (acc_norm), the standard metric for this benchmark.

Benchmark	Category	ρ	p	σ_{pure}	$\sigma_{\text{h3-1}}$	Ratio
LAMBADA‡	**Retrieval**	**−1.0**	**.017**	**2.41**	**0.49**	**4.87**
(excl. GDN)		−0.7	.233	—	—	—
BoolQ	Comprehension	0.2	.783	1.63	2.36	0.69
ARC-C	Reasoning	0.6	.350	0.96	1.32	0.72
HellaSwag	Reasoning	0.4	.517	1.17	1.22	0.96
PIQA	Reasoning	−0.5	.450	0.54	0.34	1.59
WinoGrande	Reasoning	0.1	.950	0.67	0.78	0.86
ARC-E	Mixed	−0.7	.233	1.00	0.13	7.70
MMLU	Mixed	−0.7	.233	0.92	0.41	2.27
SciQ	Mixed	0.1	.950	0.87	0.74	1.19
XStoryCloze	Mixed	−0.6	.350	0.82	0.55	1.49
COPA†	Reasoning	−0.67	.267	2.63	0.47	5.57

†Supplementary ($n = 100$); excluded from multiple comparison correction.

‡GDN h3-1 uses 16 attention heads; all other configurations use 32 (§4.3).

Bonferroni-adjusted $p = .017 \times 10 = .170$ (all-10-benchmark correction). **Power is limited: only $|\rho| = 1.0$ achieves $p < .05$ at $n = 5$.**

net 153 examples (462 helped, 615 hurt), RetNet loses 69 (426/495), and HGRN2 gains 153 (548/395). HGRN2 is the only architecture for which adding attention yields a net per-example benefit. Pairwise Jaccard similarity is low: mean $\bar{J} \leq 0.10$, indicating ∼90% non-overlap. Overlap significantly exceeds the independence baseline ($J_{\text{null}} \approx 0.05$, all $z > 5.9$, $p < .0001$; Appendix K), but normalizing by the theoretical upper bound $J_{\max} = \min(|A|, |B|)/\max(|A|, |B|)$, observed overlap reaches only 6–7% of the achievable range above chance (normalized ES 0.065 helped, 0.062 hurt; per-pair details in Appendix I), confirming that low Jaccard cannot be attributed to set-size constraints.

Confidence margin predictor. Pure-model confidence on the target token predicts which examples will be affected (AUC 0.70–0.74; Appendix P): rescue thresholds are architecture-invariant (1% spread) but disruption thresholds vary 47%, explaining cross-architecture Jaccard divergence (Appendix P.2). Calibration differences account for less than half the threshold spread (Appendix Q). The rescue pattern generalizes across three benchmarks ($d = 0.13$–0.60; Appendix P.1); disruption is weaker on reasoning tasks (Appendix R). This structured disagreement constitutes predictive multiplicity (Marx et al., 2020) under architectural variation (cf. D'Amour et al., 2022).

Confidence-based routing. As an illustrative case study, the rescue-threshold asymmetry suggests a practical routing strategy: for each example, select the architecture whose pure-model confidence best predicts successful rescue. On LAMBADA, rescue-threshold routing achieves 56.2% accuracy versus 50.9% for majority vote (+5.3 pp), extracting 36% of the oracle gap (oracle = 65.5%). On ARC-Easy, gains are marginal (+1.0 pp); on HellaSwag and PIQA, routing provides no improvement. Using 10-repeat 50/50 cross-validated splits, routing works when multiplicity is high ($\alpha = 0.30$, LAMBADA) and a clear rescue-threshold signal exists, but fails when multiplicity is low ($\alpha = 0.14$, PIQA).

Multiplicity variation. Extending the per-example analysis to HellaSwag ($N = 10{,}042$) and PIQA ($N = 1{,}838$) reveals that predictive multiplicity varies systematically across tasks (Table 2). The ambiguity α (Marx et al., 2020) ranges from 0.14 (PIQA) through 0.15 (HellaSwag) and 0.21 (ARC-Easy) to 0.30 (LAMBADA), forming a task-dependent variation. Format does not drive multiplicity: HellaSwag (four-way) and PIQA (two-choice) produce nearly identical α despite

Table 2: Predictive multiplicity at 1.3B. α: ambiguity at h3-1 (Marx et al., 2020). $\bar{J}$: mean pairwise Jaccard of helped (h) and hurt (u) sets, averaged across four hybrid attention ratios and three architecture pairs. ρ_α: Spearman ρ between attention ratio and α. Benchmarks ordered by α.

	N	α	$\bar{J}_h$	$\bar{J}_u$	ρ_α
PIQA	1838	.14	.091	.094	0.0
HellaSwag	10042	.15	.095	.085	0.1
ARC-Easy	2376	.21	.092	.099	-0.2
LAMBADA	5153	.30	.096	.091	-0.6

Within-benchmark α variation < 0.03 across ratios. BoolQ, WinoGrande: Appendix R, S.

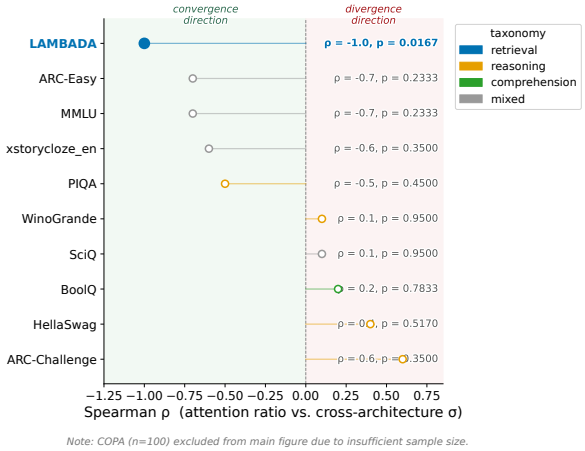


Figure 2: Spearman ρ between attention ratio and cross-architecture σ for each benchmark at 1.3B scale. LAMBADA ($\rho = -1.0$; BCa 95% CI $[-1.0, -0.7]$, heuristic) is the only benchmark with a negative correlation whose confidence interval excludes zero. Benchmarks are grouped by task category. Bootstrap CI is heuristic given $n = 3$ architectures.

different formats. Bootstrap 95% CIs confirm the ordering (all non-overlapping except PIQA–HellaSwag; Appendix, Table 12). Moreover, α does not significantly correlate with task accuracy (Spearman $\rho = -0.60$, $p = .24$). Pairwise Jaccard remains uniformly low ($\bar{J} \leq 0.10$, all $p < .01$ across four benchmarks), indicating that architectures affect different examples regardless of how many examples are affected. Within each benchmark, α is stable across attention ratios (variation < 0.03), but differs $2\times$ across task types (from reasoning to retrieval), confirming that multiplicity magnitude depends on task, not attention dose (cf. Gorbett and Jana, 2026). PIQA (0.14) and HellaSwag (0.15) share a reasoning-task floor, while retrieval (LAMBADA, 0.30) and mixed (ARC-Easy, 0.21) tasks produce higher values.

4.2 Decoupling

The per-example findings above raise a natural question: when aggregate metrics converge, does this reflect agreement on the same examples? Table 1 provides context. Only LAMBADA shows monotonic convergence ($\rho = -1.0$, Bonferroni-adjusted $p = .170$; excluding GDN, whose h3-1 uses 16 rather than 32 attention heads: $\rho = -0.7$, $p = .233$; observed at 1.3B, absent at 340M; Appendix B, Figure 2). This is a suggestive but underpowered secondary observation ($n = 5$; the test has negligible power for $|\rho| < 1.0$). Given the limited statistical power, non-significant correlations on other benchmarks should not be interpreted as absence of association. LAMBADA relies purely on in-context retrieval without external knowledge or multi-step reasoning, making it the task most directly aligned with attention's role in context access—a plausible explanation for why convergence is strongest here.

Even on LAMBADA, the answer is no. Aggregate convergence and per-example divergence operate on different timescales as attention ratio increases (Figure 1).

Cross-architecture σ decreases from 0.024 (pure) to 0.005 (h3-1), while pairwise Jaccard saturates at the first attention level ($\bar{J} \approx 0.095$) with no further increase—indicating that each architecture's per-example response to attention is largely idiosyncratic. The observed $\bar{J}$ reaches only 6–7% of the achievable range above chance ($J_{\max} \approx 0.78$–0.92), ruling out ceiling constraints (§4.1).

This saturation generalizes beyond LAMBADA: on all four main benchmarks, mean pairwise Jaccard shows no significant trend with attention ratio (all $\rho \leq 0.80$, $p \geq 0.333$; Appendix N), with values uniformly between 0.08 and 0.11.

Bootstrap 95% CIs confirm non-overlapping σ between pure and h3-1 but completely overlapping Jaccard across all hybrid ratios (§3.5). Within-architecture Jaccard ($\sim$0.33; Appendix M) is $3\times$ the cross-architecture value ($\sim$0.10); Pythia seed calibration shows a similar $3{-}4\times$ gap (Appendix W), consistent with architecture-driven divergence, though matched multi-seed experiments would be needed for definitive evidence.

Attention weight analysis (Appendix F) shows GDN and RetNet produce nearly identical attention distributions (cosine 0.995) yet differ in per-example effects, suggesting these originate in the

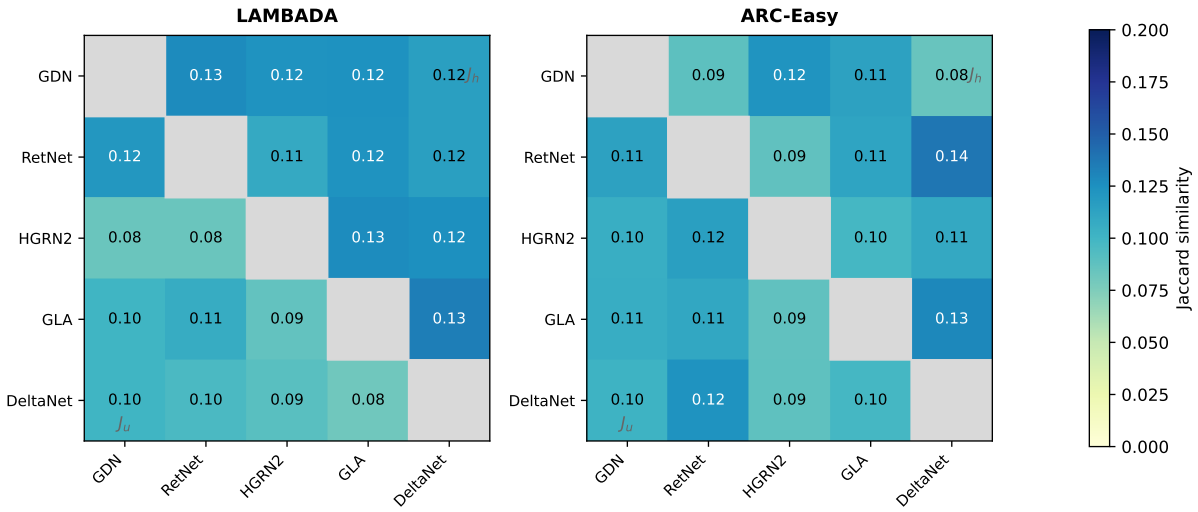


Figure 3: Pairwise Jaccard heatmap across five architectures (340M, h3-1), where helped (hurt) examples are those correct only with (without) attention. Upper triangle: J_{helped}; lower triangle: J_{hurt}. All values ≤ 0.14, all permutation tests $p \leq .002$. HellaSwag and PIQA results in Appendix X.

recurrent layers. The four-benchmark variation (Table 2) confirms independence: LAMBADA ($\alpha = 0.30$) is the only benchmark showing convergence despite having the highest multiplicity, while PIQA ($\alpha = 0.14$) and HellaSwag ($\alpha = 0.15$) show no convergence with the lowest multiplicity.

Cross-scale analysis confirms scale-robustness: pairwise Jaccard is stable ($\bar{J} = 0.08$–0.10 at both scales), though α is systematically higher at 340M ($\Delta \leq 8$ pp). At 340M, aggregate convergence is absent ($\rho = -0.50$, $p = .45$), suggesting that the *pattern* of multiplicity reflects structural differences while absolute magnitude is modulated by capacity (Appendix T).

4.3 Robustness

Removing GDN weakens LAMBADA convergence ($\rho = -0.7$, $p = .233$) and reverses MMLU's ($\rho = +0.6$), both driven by GDN's 16-head configuration (vs. 32 in others; Appendix E). The RetNet–HGRN2 pair preserves decoupling ($\sigma \leq 0.008$, $J = 0.086$–0.108). The aggregate convergence result is therefore dependent on GDN inclusion; the decoupling result is not. Five-architecture validation (GLA, DeltaNet added) confirms $J \leq 0.14$ across all ten pairs and four benchmarks (all $p < .01$; Figure 3; Appendix X). Full sensitivity analyses appear in Appendix D.

5 Conclusion

Per-example multiplicity in hybrid recurrent models varies by task ($\alpha = 0.14$–0.30), with Jaccard uniformly low (≤ 0.10) even as aggregate metrics converge on LAMBADA (dependent on GDN inclusion; §4.3); five-architecture validation confirms this across four benchmarks (all $J \leq 0.14$, $p < .01$; Appendix X). Pure-model confidence partially predicts affected examples (AUC 0.70–0.74; §4.1).

For practitioners, three implications follow from the ∼90% non-overlap: (1) $\alpha > 0.2$ warrants per-example audit before architecture substitution, as multiplicity at this level means >20% of examples change outcome; (2) low-Jaccard pairs maximize ensemble complementarity—majority vote fails despite 15–30% disagreement, but oracle accuracy reveals +7–14 pp of untapped gain; (3) rescue-threshold routing is viable when α is high and disruption thresholds differ across architectures, extracting 36% of oracle complementarity on LAMBADA (+5.3 pp; §4.1). Deployment-ready routing remains open.

Limitations

Single training runs. Each configuration corresponds to a single training run, so we cannot fully disentangle architectural from seed-specific effects. The 3× within- vs. cross-architecture Jaccard gap and Pythia 70M seed calibration (within-seed J is 2.7–4.3× higher; Appendix W) provide proxy evidence, and the consistency of significant Jaccard across all 40 architecture–benchmark combinations (all $p < .01$) would be unlikely under seed noise alone. With $n = 5$ attention ratios, the minimum

achievable p is 0.017, so LAMBADA's Bonferroni-adjusted $p = .170$ does not reach conventional significance.

Coverage. We examine three primary architectures (plus two extended) from one model zoo, four main benchmarks, and two scale points. Generalization to Mamba, RWKV, other hybrid families, and generation tasks remains open.

Pre-training only. All evaluations use pretrained checkpoints without fine-tuning; per-example effects may not persist after task-specific adaptation.

Code and Data Availability

All models are from the publicly available model zoo of Wang et al. (2025). Our analysis code and per-example evaluation outputs will be released upon publication.

References

Ameen Ali, Itamar Zimerman, and Lior Wolf. 2025. The hidden attention of mamba models. In *Proceedings of ACL*.

Simran Arora, Sabri Eyuboglu, Aman Timalsina, Isys Johnson, Michael Poli, James Zou, Atri Rudra, and Christopher Ré. 2024a. Zoology: Measuring and improving recall in efficient language models. In *Proceedings of ICLR*.

Simran Arora, Sabri Eyuboglu, Michael Zhang, Aman Timalsina, Frederic Alberti, James Bornschein, and 1 others. 2024b. Simple linear attention language models balance the recall-throughput tradeoff. In *Proceedings of ICML*.

Abhinaba Basu. 2026. When does content-based routing work? representation requirements for selective attention in hybrid sequence models. *arXiv preprint arXiv:2603.20997*.

Stella Biderman, Hailey Schoelkopf, Quentin Gregory Anthony, Herbie Bradley, Kyle O'Brien, Eric Hallahan, Mohammad Aflah Khan, Shivanshu Purohit, USVSN Sai Prashanth, Edward Raff, Aviya Skowron, Lintang Sutawika, and Oskar van der Wal. 2023. Pythia: A suite for analyzing large language models across training and scaling. In *Proceedings of ICML*.

Emily Black, Manish Raghavan, and Solon Barocas. 2022. Model multiplicity: Opportunities, concerns, and solutions. In *Proceedings of the ACM Conference on Fairness, Accountability, and Transparency (FAccT)*.

Aleksandar Botev, Soham De, Samuel L. Smith, and 1 others. 2024. Recurrentgemma: Moving past transformers for efficient open language models. *arXiv preprint arXiv:2404.07839*.

Gavin Brown, Jeremy Wyatt, Rachel Harris, and Xin Yao. 2005. Diversity creation methods: A survey and categorisation. *Information Fusion*, 6(1):5–20.

Peter Clark, Isaac Cowhey, Oren Etzioni, Tushar Khot, Ashish Sabharwal, Carissa Schoenick, and Oyvind Tafjord. 2018. Think you have solved question answering? try ARC, the AI2 reasoning challenge. *arXiv preprint arXiv:1803.05457*.

A. Feder Cooper, Katherine Abrams, and Solon Barocas. 2024. Arbitrariness and social prediction: The confounding role of variance in fair classification. In *Proceedings of AAAI*.

Alexander D'Amour, Katherine Heller, Dan Moldovan, and 1 others. 2022. Underspecification presents challenges for credibility in modern machine learning. *Journal of Machine Learning Research*, 23(226):1–61.

Tri Dao and Albert Gu. 2024. Transformers are SSMs: Generalized models and efficient algorithms through structured state space duality. In *Proceedings of ICML*.

Soham De, Samuel L. Smith, Anushan Fernando, and 1 others. 2024. Griffin: Mixing gated linear recurrences with local attention for efficient language models. *arXiv preprint arXiv:2402.19427*.

Jesse Dodge, Gabriel Ilharco, Roy Schwartz, Ali Farhadi, Hannaneh Hajishirzi, and Noah A. Smith. 2020. Fine-tuning pretrained language models: Weight initializations, data orders, and early stopping. *arXiv preprint arXiv:2002.06305*.

Xin Dong and 1 others. 2024. Hymba: A hybrid-head architecture for small language models. *arXiv preprint arXiv:2411.13676*.

Ally Yalei Du, Dung Daniel Ngo, and Zhiwei Steven Wu. 2025. Reconciling model multiplicity for downstream decision making. In *Proceedings of ICLR*.

Finlay Fehlauer, Kyle Mahowald, and Tiago Pimentel. 2025. Convergence and divergence of language models under different random seeds. In *Proceedings of EMNLP*, pages 32982–32991.

Aaron Fisher, Cynthia Rudin, and Francesca Dominici. 2019. All models are wrong, but many are useful: Learning a variable's importance by studying an entire class of prediction models simultaneously. *Journal of Machine Learning Research*, 20(177):1–81.

Stanislav Fort, Huiyi Hu, and Balaji Lakshminarayanan. 2019. Deep ensembles: A loss landscape perspective. *arXiv preprint arXiv:1912.02757*.

Daniel Y. Fu, Tri Dao, Khaled K. Saab, Armin W. Thomas, Atri Rudra, and Christopher Ré. 2023. Hungry hungry hippos: Towards language modeling with state space models. In *Proceedings of ICLR*.

Leo Gao, Jonathan Tow, Baber Abbasi, Stella Biderman, and 1 others. 2024. A framework for few-shot language model evaluation. Zenodo.

Matt Gorbett and Suman Jana. 2026. Cross-model disagreement as a label-free correctness signal. *arXiv preprint arXiv:2603.25450*.

Albert Gu and Tri Dao. 2024. Mamba: Linear-time sequence modeling with selective state spaces. *arXiv preprint arXiv:2312.00752*.

Albert Gu, Karan Goel, and Christopher Ré. 2022. Efficiently modeling long sequences with structured state spaces. In *Proceedings of ICLR*.

Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q. Weinberger. 2017. On calibration of modern neural networks. In *Proceedings of ICML*, pages 1321–1330.

Parian Haghighat, Hadis Anahideh, and Cynthia Rudin. 2026. Resolving predictive multiplicity for the rashomon set. *arXiv preprint arXiv:2601.09071*.

Hsiang Hsu and Flavio P. Calmon. 2022. Rashomon capacity: A metric for predictive multiplicity in classification. In *Advances in Neural Information Processing Systems*.

Suyash Jain, Yuanjun Wang, Kathleen Creel, and Christo Wilson. 2025. Allocation multiplicity: Evaluating the promises of the Rashomon set. In *Proceedings of the 2025 ACM Conference on Fairness, Accountability, and Transparency (FAccT)*.

Angelos Katharopoulos, Apoorv Vyas, Nikolaos Pappas, and François Fleuret. 2020. Transformers are RNNs: Fast autoregressive transformers with linear attention. In *Proceedings of ICML*.

Ludmila I. Kuncheva and Christopher J. Whitaker. 2003. Measures of diversity in classifier ensembles and their relationship with the ensemble accuracy. *Machine Learning*, 51(2):181–207.

Aakash Lahoti, Kevin Y. Li, Berlin Chen, Caitlin Wang, Aviv Bick, J. Zico Kolter, Tri Dao, and Albert Gu. 2026. Mamba-3: Improved sequence modeling using state space principles. In *Proceedings of ICLR*.

Opher Lieber, Barak Lenz, Hofit Bata, and 1 others. 2024. Jamba: A hybrid transformer-mamba language model. *arXiv preprint arXiv:2403.19887*.

Charles Marx, Flavio P. Calmon, and Berk Ustun. 2020. Predictive multiplicity in classification. In *Proceedings of ICML*.

R. Thomas McCoy, Junghyun Min, and Tal Linzen. 2020. BERTs of a feather do not generalize together: Large variability in generalization across models with similar test set performance. In *Proceedings of NeurIPS*.

Mahdi Milani Fard, Kevin Canini, Andrew Cotter, Jan Pfeifer, and Peter Bartlett. 2016. Launch and iterate: Reducing prediction churn. In *Advances in Neural Information Processing Systems*.

Timothy Niven and Hung-Yu Kao. 2019. Probing neural network comprehension of natural language arguments. In *Proceedings of ACL*.

Denis Paperno, Germán Kruszewski, Angeliki Lazaridou, and 1 others. 2016. The LAMBADA dataset: Word prediction requiring a broad discourse context. In *Proceedings of ACL*.

Bo Peng and 1 others. 2024. Eagle and finch: RWKV with matrix-valued states and dynamic recurrence. *arXiv preprint arXiv:2404.05892*.

Michael Poli, Stefano Massaroli, Eric Nguyen, Daniel Y. Fu, Tri Dao, Stephen Baccus, Yoshua Bengio, Stefano Ermon, and Christopher Ré. 2023. Hyena hierarchy: Towards larger convolutional language models. In *Proceedings of ICML*.

Zhen Qin, Songlin Li, Weixuan Zhong, Yiran Sun, Dong Li, Hao Peng, Zhijun Yang, and Sheng Zhong. 2023. Hierarchically gated recurrent neural network for sequence modeling. In *Advances in Neural Information Processing Systems*.

Zhen Qin and 1 others. 2024. HGRN2: Gated linear RNNs with state expansion. *arXiv preprint arXiv:2404.07904*.

Liliang Ren, Yang Liu, Yadong Yan, Yi Zhu, and 1 others. 2024. Samba: Simple hybrid state space models for efficient unlimited context language modeling. *arXiv preprint arXiv:2406.07522*.

William S. Robinson. 1950. Ecological correlations and the behavior of individuals. *American Sociological Review*, 15(3):351–357.

Thibault Sellam, Steve Yadlowsky, Ian Tenney, Jason Wei, Naomi Saphra, Alexander D'Amour, Tal Linzen, Jasmijn Bastings, Iulia Raluca Turc, Jacob Eisenstein, Dipanjan Das, and Ellie Pavlick. 2022. The MultiBERTs: BERT reproductions for robustness analysis. In *Proceedings of ICLR*.

Lesia Semenova, Cynthia Rudin, and Ronald Parr. 2022. On the existence of simpler machine learning models. In *Proceedings of FAccT*, pages 1827–1858.

Yutao Sun, Li Dong, Shaohan Huang, Shuming Ma, Yuqing Xia, Jilong Xue, Jianyong Wang, and Furu Wei. 2023. Retentive network: A successor to transformer for large language models. *arXiv preprint arXiv:2307.08621*.

Swabha Swayamdipta, Roy Chandrasekaran, John Wieting, Hannaneh Hajishirzi, and Noah A. Smith. 2020. Dataset cartography: Mapping and diagnosing datasets with training dynamics. In *Proceedings of EMNLP*.

Oskar van der Wal, Pietro Lesci, Max Müller-Eberstein, Naomi Saphra, Hailey Schoelkopf, Willem Zuidema, and Stella Biderman. 2025. Polypythias: Stability and outliers across fifty language model pre-training runs. In *Proceedings of ICLR*.

Roger Waleffe, Wonmin Byeon, Duncan Riach, Brandon Norick, and 1 others. 2024. An empirical study of mamba-based language models. *arXiv preprint arXiv:2406.07887*.

Dustin Wang, Rui-Jie Zhu, Angelos Katharopoulos, and Albert Gu. 2025. A systematic analysis of hybrid linear attention. *arXiv preprint arXiv:2507.06457*.

Jamelle Watson-Daniels, David C. Parkes, and Berk Ustun. 2023. Predictive multiplicity in probabilistic classification. In *Proceedings of AAAI*, volume 37, pages 10306–10314.

Songlin Yang, Jan Kautz, and Ali Hatamizadeh. 2024. Gated delta networks: Improving mamba2 with delta rule. *arXiv preprint arXiv:2412.06464*.

A Confirmatory and Exploratory Analyses

The task taxonomy used in §3.2 was defined prior to running any convergence analyses. Each benchmark was assigned to exactly one of four categories based on the primary cognitive demand of the task:

- **Retrieval**: tasks requiring prediction from long-range context without multi-step reasoning (LAMBADA).
- **Comprehension**: reading comprehension with explicit passage support (BoolQ).
- **Reasoning**: tasks requiring commonsense inference or multi-step reasoning (HellaSwag, ARC-Challenge, WinoGrande, PIQA).
- **Mixed**: tasks combining retrieval, comprehension, and reasoning demands (ARC-Easy, MMLU, SciQ, XStoryCloze).

Confirmatory benchmark set. Six benchmarks were designated as confirmatory for Bonferroni correction: LAMBADA, HellaSwag, ARC-Challenge, WinoGrande, ARC-Easy, and MMLU. These were selected as representative of retrieval, reasoning, and mixed categories before any convergence analysis was performed. The taxonomy and benchmark selection were fixed before analysis as an informal pre-analysis plan; we describe this as a confirmatory/exploratory split rather than a formal pre-registration, since no external timestamp exists. The main text applies the more conservative all-10-benchmark correction (Bonferroni-adjusted $p = .017 \times 10 = .170$). **Sensitivity:** correcting across only the six confirmatory benchmarks yields adjusted $p = .017 \times 6 = .100$, still above conventional significance.

One-sided test. If one adopts the prior that attention should reduce cross-architecture variance, the one-sided permutation p-value for LAMBADA is $1/120 \approx 0.0083$. Under 10-benchmark Bonferroni correction, this yields an adjusted $p = 0.083$, above the conventional threshold. We report two-sided tests throughout the main text as the more conservative choice; the one-sided result is noted here for completeness.

Exploratory benchmarks. Four additional benchmarks (BoolQ, PIQA, SciQ, XStoryCloze) were added as exploratory tests. All benchmarks share the same 10-benchmark correction (adjusted threshold $\alpha/10 = 0.005$) and the four exploratory benchmarks are labeled as such throughout the paper.

Supplementary benchmark. COPA ($n = 100$) is treated as supplementary due to its small sample size. It is excluded from both correction families and reported separately with the † annotation in Table 1.

B Convergence Detail

Beyond LAMBADA ($\rho = -1.0$), two benchmarks show moderate negative correlations that do not reach significance: ARC-Easy and MMLU (both $\rho = -0.7$, $p = .233$). Both are classified as "Mixed" tasks combining retrieval with reasoning demands. The sensitivity analysis (Appendix D) shows that MMLU's correlation reverses upon GDN exclusion ($\rho = +0.6$), indicating dependence on a single architecture; ARC-Easy's correlation is more stable. COPA ($n = 100$; supplementary) shows $\rho = -0.67$ with a large σ ratio (5.57), consistent with LAMBADA's pattern but unreliable due to small sample size.

Several benchmarks show positive or near-zero ρ (BoolQ: $+0.2$; ARC-C: $+0.6$; HellaSwag: $+0.4$; WinoGrande: $+0.1$; SciQ: $+0.1$), indicating that attention does not systematically reduce cross-architecture variance on reasoning or comprehension tasks. The contrast with LAMBADA is consistent with attention's convergence benefit being specific to tasks requiring direct token-level recall (Arora et al., 2024a,b) rather than multi-step inference.

Regression to the mean. The direction of LAMBADA's aggregate convergence is consistent with regression to the mean: GDN (highest pure accuracy, 0.543) declines to 0.513 under h3-1, while

HGRN2 (lowest, 0.484) rises to 0.514, yielding a $5\times$ σ compression ($0.024 \rightarrow 0.005$). However, three observations argue against a purely statistical artifact. First, the $5\times$ reduction in cross-architecture σ far exceeds typical regression amplitudes. Second, the direction is entirely determined by architecture identity, not random fluctuation: HGRN2 accuracy increases monotonically across all attention ratios, while GDN decreases monotonically—a pattern inconsistent with noise-driven mean reversion. Third, per-example analysis (§4.1) reveals architecture-specific patterns: GDN's net -153 examples and HGRN2's net $+153$ examples reflect distinct helped/hurt profiles with low cross-architecture Jaccard overlap, indicating that the aggregate convergence arises from structured, architecture-dependent per-example changes rather than symmetric regression.

C Scale-Dependent Convergence

At 1.3B parameters, LAMBADA convergence is monotonic: σ decreases at every attention level from 0% through 25% ($\rho = -1.0$). At 340M, convergence is partial: σ reaches a maximum reduction of 56% at intermediate attention ratios but rebounds at h3-1, yielding a net endpoint reduction of 36%. The trajectory is non-monotonic, and the highest-attention model does not achieve the lowest cross-architecture variance. One possible interpretation is that at smaller scale, the capacity overhead of attention layers creates interference that offsets the retrieval benefit.

This contrast is a preliminary observation from two data points (not a scaling law). Despite the aggregate divergence at 340M, mean pairwise Jaccard is similar at both scales ($J \approx 0.09$), extending the decoupling observation (§4.2) to the scale dimension.

D Sensitivity Analyses

D.1 Bootstrap Confidence Intervals

We construct 95% bootstrap confidence intervals for the Spearman ρ between attention ratio and cross-architecture σ on LAMBADA using 10,000 resamples at the architecture level ($n = 3$ architectures). Of these resamples, 11.25% are degenerate: they consist of same-architecture triplicates, which produce a constant σ vector and therefore an undefined ρ (division by zero in the rank correlation formula). These are excluded from the confidence interval computation because they carry no information about the correlation, not because they represent $\rho = 0$.

The BCa (bias-corrected and accelerated) correction degenerates at the $\rho = -1.0$ boundary because the acceleration parameter requires finite jackknife variance, which collapses when the point estimate is at the distributional limit. We therefore report percentile confidence intervals instead. From the 88.75% informative resamples, the 95% percentile CI is $[-1.0, -0.7]$, which excludes zero.

D.2 GDN Exclusion Sensitivity

The GDN h3-1 configuration uses 16 attention heads rather than 32 (see Appendix E). To assess the influence of this confound, we recompute all convergence correlations after excluding GDN, leaving two architectures (RetNet, HGRN2) at five attention ratios.

Table 3 reports the results for benchmarks where the correlation changes by $|\Delta\rho| \geq 0.3$. LAMBADA weakens from $\rho = -1.0$ to $\rho = -0.7$ ($p = .233$), consistent with GDN contributing to the convergence signal. MMLU reverses from $\rho = -0.7$ to $\rho = +0.6$, confirming that the moderate negative correlation on MMLU depends on GDN's inclusion and should not be interpreted as evidence of convergence.

Table 3: Effect of GDN exclusion on Spearman ρ (attention ratio vs. cross-architecture σ) at 1.3B scale. Only benchmarks with $|\Delta\rho| \geq 0.3$ shown.

Benchmark	ρ_{all}	$\rho_{\text{no-GDN}}$	$\Delta\rho$	$p_{\text{no-GDN}}$
LAMBADA	-1.0	-0.7	$+0.3$	.233
MMLU	-0.7	$+0.6$	$+1.3$	.350

E GDN Attention Head Confound

In the Wang et al. (2025) model zoo, the GDN h3-1 configuration at 1.3B scale uses 16 attention heads per attention layer, while all other hybrid configurations (across GDN, RetNet, and HGRN2) use 32 attention heads. This is a design choice in the original model zoo, not an error. The reduced head count changes the per-head dimension and may affect the attention capacity of GDN h3-1 relative to other configurations.

This confound is relevant because the convergence analysis measures cross-architecture σ at each attention ratio. At the h3-1 ratio (25% attention), GDN's attention layers have a different internal structure than those of RetNet and HGRN2,

which could contribute to variance reduction (or inflation) for reasons unrelated to the recurrent–attention interaction under study.

The sensitivity analysis in Appendix D shows that removing GDN weakens the LAMBADA convergence correlation from $\rho = -1.0$ to $\rho = -0.7$. This weakening is partially attributable to the reduced sample size ($n = 2$ architectures) and partially to the confound itself. We cannot separate these two effects with the available data.

F Attention Weight Analysis

We extract attention weight matrices from the 1.3B h3-1 models of all three architectures using forward hooks on each attention layer. While Ali et al. (2025) show that SSM layers learn implicit attention patterns, our analysis focuses on the explicit attention layers interleaved within the hybrid architecture. For each layer, we compute the mean attention distribution across all 5,153 LAMBADA examples and measure cross-architecture similarity via cosine similarity.

F.1 Cross-Architecture Cosine Similarity

Table 4 reports cosine similarity between mean attention distributions at selected layers and averaged across all layers.

Table 4: Cosine similarity between mean attention weight distributions of architecture pairs at selected layers (1.3B h3-1 models on LAMBADA).

Layer	GDN–RetNet	GDN–HGRN2	HGRN2–RetNet
L2 (early)	0.995	0.244	0.196
L20 (deep)	0.986	0.942	0.943
Mean (all layers)	0.905	0.766	0.698

At layer 2, GDN and RetNet produce nearly identical attention distributions (cosine 0.995), while HGRN2 diverges from both (cosine < 0.25). By layer 20, all three architectures converge to similar attention patterns (cosine > 0.94). The layer-wise trajectory suggests that early-layer attention patterns are shaped by the preceding recurrent layers (which differ across architectures), while deeper layers converge regardless of the recurrent substrate.

F.2 Helped vs. Hurt Attention Entropy

To test whether the attention mechanism itself differs between helped and hurt examples, we compute the Shannon entropy of the attention distribution for each example at each attention layer and compare the mean entropy of helped examples against hurt examples within each architecture.

Table 5: Mean attention entropy difference (helped − hurt) averaged across attention layers (1.3B h3-1 models on LAMBADA). Values near zero indicate that the attention mechanism does not discriminate between helped and hurt examples.

Architecture	ΔH (helped − hurt)
GDN	-0.0009
RetNet	$+0.0179$
HGRN2	$+0.0020$

GDN and HGRN2 show negligible entropy differences ($|\Delta H| \leq 0.002$). RetNet's $\Delta H = +0.018$ is small in absolute terms but slightly larger; helped examples receive marginally more diffuse attention in RetNet. Across all three architectures, the differences are insufficient to explain the per-example divergence, consistent with per-example effects not being explained by attention weight patterns and likely originating in the recurrent layers.

F.3 Last-5 Attention Mass

We also measure the fraction of attention mass allocated to the last five tokens, which captures whether the attention mechanism concentrates differently on recent context for helped vs. hurt examples.

Table 6: Mean attention mass on last-5 tokens for helped and hurt examples, averaged across attention layers (1.3B h3-1 models on LAMBADA).

Architecture	Helped	Hurt
GDN	0.0834	0.0831
RetNet	0.1578	0.1591
HGRN2	0.0453	0.0467

No architecture shows a meaningful difference in last-5 attention mass between helped and hurt examples (Table 6). This further supports the conclusion that the per-example divergence is not driven by differential attention allocation.

G Cross-Scale Per-Example Comparisons

G.1 Within-Scale Pairwise Jaccard

Table 7 reports the within-scale pairwise Jaccard similarity between architectures' helped and hurt sets at both scales. Mean values are similar across scales (340M: 0.094/0.090; 1.3B: 0.100/0.093),

confirming that per-example architecture specificity is approximately scale-robust.

Table 7: Within-scale pairwise Jaccard similarity of per-example effects (pure → h3-1) on LAMBADA. Each row compares the helped (or hurt) sets of two architectures at the same scale.

Scale	Pair	J_{helped}	J_{hurt}
340M	GDN–RetNet	0.092	0.109
	GDN–HGRN2	0.084	0.075
	RetNet–HGRN2	0.108	0.085
	Mean	*0.094*	*0.090*
1.3B	GDN–RetNet	0.092	0.103
	GDN–HGRN2	0.100	0.089
	RetNet–HGRN2	0.108	0.087
	Mean	*0.100*	*0.093*

G.2 Cross-Scale Pairwise Jaccard

For each architecture, we compare the helped and hurt sets of the pure → h3-1 transition at 340M and 1.3B scale on LAMBADA. The comparison uses the same classification scheme as §4.1: an example is "helped" if it is correct only in the hybrid variant and "hurt" if correct only in the pure baseline. All pairwise Jaccard values fall below 0.10, indicating that scale produces per-example predictive multiplicity analogous to the cross-architecture effect reported in §4.1.

Table 8: Cross-scale pairwise Jaccard similarity of per-example effects (pure → h3-1) between 340M and 1.3B on LAMBADA. All values below 0.10.

Architecture	Scale Pair	J_{helped}	J_{hurt}
GDN	340M ↔ 1.3B	0.047	0.087
RetNet	340M ↔ 1.3B	0.059	0.073
HGRN2	340M ↔ 1.3B	0.089	0.055
Mean		0.065	0.072

H Per-Example Effect Counts

Table 9 reports the full per-example effect counts for each architecture at 1.3B scale.

I Normalized Jaccard Effect Size

To assess whether low observed Jaccard reflects a floor effect imposed by unequal set sizes, we compute the theoretical upper bound $J_{\max} = \min(|A|, |B|)/\max(|A|, |B|)$ and the normalized effect size $(J_{\text{obs}} - J_{\text{null}})/(J_{\max} - J_{\text{null}})$, which measures observed overlap as a fraction of the achievable range above chance. Table 10 reports per-pair results.

Table 9: Per-example effects of attention (pure → h3-1) on 5,153 LAMBADA examples at 1.3B scale. $\bar{J}$: mean pairwise Jaccard similarity with other architectures across helped and hurt sets.

Architecture	Helped	Hurt	Net	$\bar{J}$
GDN	462	615	−153	.096
RetNet	426	495	−69	.100
HGRN2	548	395	+153	.104

Table 10: Normalized Jaccard effect size for each architecture pair (1.3B h3-1 on LAMBADA). $J_{\max}$: theoretical upper bound given set sizes. Normalized ES: fraction of achievable overlap above chance.

Category	Pair	J_{obs}	$J_{\max}$	J_{null}	Norm. ES
Helped	GDN–RetNet	.086	.922	.048	.044
	GDN–HGRN2	.099	.843	.050	.062
	RetNet–HGRN2	.108	.777	.042	.090
	Mean	*.098*	*.847*	*.047*	*.065*
Hurt	GDN–RetNet	.092	.805	.056	.048
	GDN–HGRN2	.070	.642	.043	.045
	RetNet–HGRN2	.087	.798	.050	.049
	Mean	*.083*	*.748*	*.050*	*.047*

All normalized effect sizes fall below 0.10, confirming that observed overlap reaches only 6–7% of the theoretical maximum. The low Jaccard values cannot be attributed to set-size constraints.

J ARC-Easy Per-Example Analysis

Table 11 reports per-example effect counts on ARC-Easy ($N = 2{,}376$) at 1.3B scale for each attention ratio. Unlike LAMBADA, ARC-Easy shows no aggregate convergence ($\rho = -0.20$, $p = .783$). HGRN2 is net positive at all ratios; GDN and RetNet are mixed.

Table 12 reports the ambiguity α (Marx et al., 2020) at each attention ratio for all four main benchmarks. Within-benchmark ranges are 0.004 (HellaSwag), 0.011 (PIQA), 0.025 (ARC-Easy), and 0.022 (LAMBADA), all below the 0.03 threshold reported in §4.1.

K Jaccard Permutation Test

To assess whether the observed pairwise Jaccard similarities exceed what would be expected under independence, we perform a permutation test. For each architecture pair at 1.3B h3-1, we hold the helped (or hurt) set sizes fixed and randomly assign example indices to each set (10,000 permutations,

Table 11: Per-example effects on ARC-Easy at 1.3B scale. Net = helped − hurt.

Architecture	Ratio	Helped	Hurt	Net
GDN	h24-1	178	185	−7
	h12-1	176	206	−30
	h6-1	170	157	+13
	h3-1	171	180	−9
RetNet	h24-1	158	168	−10
	h12-1	163	146	+17
	h6-1	159	147	+12
	h3-1	144	184	−40
HGRN2	h24-1	160	119	+41
	h12-1	159	111	+48
	h6-1	185	118	+67
	h3-1	164	146	+18

Table 12: Ambiguity α across attention ratios (1.3B scale). 95% CI: BCa bootstrap (10,000 resamples) at the pure-recurrent configuration.

Benchmark	Pure	h24-1	h12-1	h6-1	h3-1	95% CI (pure)
PIQA	.143	.138	.132	.143	.140	[.127, .159]
HellaSwag	.148	.150	.148	.146	.150	[.141, .155]
ARC-Easy	.205	.226	.223	.201	.210	[.189, .221]
LAMBADA	.307	.300	.316	.294	.296	[.294, .319]

seed 42). Table 13 reports the observed J, the null baseline J_{null} (mean over permutations), and the z-score.

Table 13: Permutation test for pairwise Jaccard similarity (1.3B h3-1 on LAMBADA, 10,000 permutations). All observed values significantly exceed the independence baseline.

Category	Pair	J_{obs}	J_{null}	z	p
Helped	GDN–HGRN2	.100	.051	7.11	$<.0001$
	GDN–RetNet	.092	.045	6.75	$<.0001$
	HGRN2–RetNet	.108	.049	8.61	$<.0001$
Hurt	GDN–HGRN2	.088	.049	5.91	$<.0001$
	GDN–RetNet	.103	.056	6.85	$<.0001$
	HGRN2–RetNet	.087	.045	6.05	$<.0001$

All six comparisons yield $z > 5.9$ ($p < .0001$). The null baseline averages $J_{\text{null}} \approx 0.049$; observed values are approximately twice this level. The overlap between architectures' per-example effect sets is therefore structured rather than a product of random set overlap given the set sizes.

L Transformer Entrainment Comparison

To test whether attention layers make hybrid models' per-example behavior converge toward a pure Transformer, we compare the correct-example sets of each architecture's pure and h3-1 variants against a 1.3B Transformer baseline (accuracy 48.1%) on LAMBADA. Table 14 reports the Jaccard similarity between each model's correct set and the Transformer's correct set.

Table 14: Jaccard similarity between each model's correct set and the Transformer correct set on LAMBADA (1.3B scale). All three h3-1 hybrids are more similar to the Transformer than their pure counterparts.

Architecture	$J_{\text{pure,T}}$	$J_{\text{h3-1,T}}$	Δ
GDN	0.645	0.649	+0.005
RetNet	0.679	0.694	+0.015
HGRN2	0.672	0.701	+0.029

All three architectures show positive Δ: the h3-1 correct set is more similar to the Transformer correct set than the pure correct set is. The effect is strongest for HGRN2 (+0.029) and weakest for GDN (+0.005). This ordering is consistent with HGRN2 being the only architecture with a positive net per-example effect from attention (Table 9). The Transformer itself achieves the lowest accuracy among all models, so the convergence is not explained by hybrid models degrading toward Transformer-level performance; rather, adding attention shifts which specific examples the hybrid gets correct toward the Transformer's pattern.

M Within-Architecture Per-Example Stability

Per-example effect sets exhibit a three-tier Jaccard hierarchy on LAMBADA (1.3B scale). Under independence (permutation baseline), pairwise Jaccard is $J \approx 0.05$. Cross-architecture comparisons (different architectures at the same attention ratio) yield $J \approx 0.10$, approximately $2\times$ baseline. Within-architecture comparisons (same architecture across different attention ratios) yield $J \approx 0.33$, approximately $7\times$ baseline and $3\times$ the cross-architecture level. Architecture identity therefore has approximately three times the influence of attention dose on per-example effects.

Table 15 reports mean within-architecture Jaccard for each architecture. HGRN2 shows the highest helped overlap and lowest hurt overlap, consistent with its monotonically increasing helped count across ratios (479 → 548).

Table 16 reports the full pairwise Jaccard matrix. Attention ratios are denoted by their hk-1

Table 15: Mean within-architecture pairwise Jaccard similarity across attention ratios (1.3B, LAMBADA). Each mean is computed over all six ratio pairs per architecture.

Architecture	$\bar{J}_{\text{helped}}$	$\bar{J}_{\text{hurt}}$
GDN	0.338	0.319
RetNet	0.333	0.321
HGRN2	0.373	0.291
Overall mean	*0.348*	*0.310*

labels: h24-1 (4%), h12-1 (7.7%), h6-1 (14.3%), h3-1 (25%).

Table 16: Within-architecture pairwise Jaccard similarity between helped and hurt sets at different attention ratios (1.3B, LAMBADA). Each cell compares the per-example effect sets of two attention configurations within the same architecture.

		Ratio pair					
Arch	Cat	h24–h12	h24–h6	h24–h3	h12–h6	h12–h3	h6–h3
GDN	Helped	.345	.335	.311	.344	.341	.352
	Hurt	.310	.328	.320	.331	.281	.345
RetNet	Helped	.354	.314	.334	.322	.330	.345
	Hurt	.316	.334	.307	.301	.324	.343
HGRN2	Helped	.359	.369	.375	.368	.378	.388
	Hurt	.292	.289	.298	.273	.309	.287

Adjacent ratio pairs (e.g., h24-1 vs. h12-1) do not exhibit systematically higher Jaccard than non-adjacent pairs (e.g., h24-1 vs. h3-1). Across all architectures, the mean Jaccard for adjacent pairs is 0.353 (helped) and 0.311 (hurt), compared with 0.343 (helped) and 0.310 (hurt) for non-adjacent pairs. The negligible difference indicates that the mapping from attention dose to per-example effects is non-smooth: doubling the attention ratio does not produce a proportionally more similar set of affected examples.

N Cross-Benchmark Jaccard Trajectories

To test whether Jaccard saturation is specific to LAMBADA or a general property, we compute mean pairwise Jaccard trajectories across all four hybrid attention ratios on all four main benchmarks. Table 17 reports the results. On every benchmark, Jaccard shows no statistically significant trend with attention ratio (all $\rho \leq 0.80$, $p \geq 0.333$), with values uniformly between 0.08 and 0.11. The saturation at the first attention level observed on LAMBADA (§4.2) thus generalizes: once any attention is present, each architecture's per-example response is largely fixed regardless of attention dose.

Table 17: Mean pairwise Jaccard (helped sets, 3 architecture pairs) across attention ratios and benchmarks (1.3B scale). ρ: Spearman correlation between attention ratio and $\bar{J}$. All $p \geq 0.333$.

Benchmark	h24-1	h12-1	h6-1	h3-1	ρ	p
LAMBADA	.091	.088	.096	.097	0.80	.333
HellaSwag	.084	.081	.091	.086	0.60	.417
PIQA	.089	.096	.089	.095	0.00	1.00
ARC-Easy	.084	.099	.091	.109	0.80	.333

O Continuous-Threshold Jaccard Analysis

The main text binarizes per-example effects at the correctness boundary. To verify that low cross-architecture Jaccard is not an artifact of this binarization, we compute Jaccard similarity using a continuous confidence threshold τ: two architectures' effect sets are defined as the examples whose target-token $|\log p|$ changes by more than τ when attention is added. On LAMBADA, mean pairwise Jaccard decreases monotonically from 0.26 ($\tau = 0.00$, any change) to 0.05 ($\tau = 0.30$, large changes only). ARC-Easy and PIQA show the same monotonic decrease. The binary threshold ($\tau = 0$, correctness flip) thus represents the upper end of cross-architecture overlap; restricting to confident transitions further reduces agreement, confirming that low Jaccard is robust to the choice of binarization threshold.

P Target-Token Confidence Margin

We analyze whether pure-model confidence on the target token predicts whether attention will help or hurt on a given example. We define confidence as $|\log p_{\text{pure}}(y^*)|$, where y^* is the target token: lower values indicate higher probability (near-correct). We initially tested raw perplexity as a predictor of helped/hurt status, but found that within-category controls (C1/C2) are necessary to isolate attention effects from the definitional confound between pure-correct and pure-wrong populations.

Table 18 compares hurt versus stayed-right samples (both pure-correct): hurt samples have significantly lower target-token probability (higher $|\log p|$), confirming that weakly-correct predictions are vulnerable to attention disruption. Table 19 compares helped versus stayed-wrong samples (both pure-wrong): helped samples have

higher target-token probability (lower $|\log p|$), confirming that near-miss errors are rescuable. This pattern is consistent across all three architectures ($|d| = 0.73$–0.92, all $p < 10^{-54}$) but does not alter the cross-architecture Jaccard findings—within-architecture predictability coexists with cross-architecture divergence.

Table 18: Control 1: Pure-correct samples—hurt vs. stayed-right. Higher mean $|\log p|$ indicates lower target-token probability (less confident correct).

Architecture	n_{hurt}	n_{stay}	Hurt mean	Stay mean	Cohen's d
GDN (h12-1)	519	2281	1.049	0.576	0.87
RetNet (h3-1)	497	2169	1.293	0.785	0.83
HGRN2 (h3-1)	392	2102	1.481	0.899	0.92

Table 19: Control 2: Pure-wrong samples—helped vs. stayed-wrong. Lower mean $|\log p|$ indicates higher target-token probability (near-miss error).

Architecture	n_{help}	n_{stay}	Help mean	Stay mean	Cohen's d
GDN (h12-1)	480	1873	2.459	4.189	−0.75
RetNet (h3-1)	420	2067	2.478	4.103	−0.73
HGRN2 (h3-1)	548	2111	2.500	4.112	−0.77

P.1 Cross-Benchmark Extension

The confidence margin pattern extends beyond LAMBADA to multiple-choice benchmarks, though with smaller effect sizes. Table 20 reports the rescue analysis (helped vs. stayed-wrong) on HellaSwag and ARC-Easy. All nine comparisons (three benchmarks × three architectures) are significant, confirming that attention consistently rescues near-miss errors regardless of task type. Effect sizes range from small on HellaSwag ($d = 0.13$–0.17) to medium on ARC-Easy ($d = 0.51$–0.60), compared to large on LAMBADA ($d = 0.73$–0.77); the attenuation on HellaSwag reflects noisier confidence proxies under length-normalized scoring. Table 21 reports the disruption analysis (hurt vs. stayed-right); two of six comparisons are significant (all on ARC-Easy) with small effect sizes ($d = 0.13$–0.26), indicating that the disruption signal is weaker on multiple-choice tasks and absent on HellaSwag under `acc_norm`. The asymmetry between rescue and disruption reflects task structure: near-miss errors are an intrinsic sample property (the model almost produced the correct answer regardless of format), while borderline correctness in multiple-choice settings depends on relative option ranking rather than absolute confidence, reducing the raw log-likelihood margin's predictive power for vulnerability.

Table 20: Cross-benchmark rescue analysis (Control 2): helped vs. stayed-wrong on HellaSwag and ARC-Easy (1.3B h3-1).

Benchmark	Architecture	Cohen's d	p
HellaSwag	GDN	−0.166	6.1×10^{-4}
	RetNet	−0.132	1.4×10^{-2}
	HGRN2	−0.164	1.4×10^{-3}
ARC-Easy	GDN	−0.60	1.0×10^{-13}
	RetNet	−0.51	5.3×10^{-10}
	HGRN2	−0.53	5.4×10^{-10}

Table 21: Cross-benchmark disruption analysis (Control 1): hurt vs. stayed-right on HellaSwag and ARC-Easy (1.3B h3-1).

Benchmark	Architecture	Cohen's d	p	Sig.?
HellaSwag	GDN	0.009	.999	No
	RetNet	−0.035	.453	No
	HGRN2	−0.062	.274	No
ARC-Easy	GDN	0.26	3.2×10^{-4}	Yes
	RetNet	0.22	2.8×10^{-3}	Yes
	HGRN2	0.13	.145	No

P.2 Prediction Framework

We formalize the confidence margin analysis as a binary prediction task: given a pure-model example and its target-token $|\log p|$, predict whether attention will change the example's correctness. Using the within-category summary statistics (Tables 18, 19), we derive analytic AUC from Cohen's d via $\text{AUC} = \Phi(|d|/\sqrt{2})$ and compute Gaussian-intersection thresholds (the $|\log p|$ value at which the two group densities intersect under equal priors).

Table 22 reports within-architecture prediction performance. All AUCs exceed 0.69, confirming that pure-model confidence is a reliable within-architecture predictor of attention effects. Disruption prediction is consistently stronger than rescue (mean AUC 0.73 vs. 0.70), reflecting the larger effect sizes in Control 1.

Despite similar within-architecture AUC, the optimal prediction thresholds differ across architectures for disruption but not rescue (Table 23). Rescue thresholds cluster tightly (range 0.03 in $|\log p|$), indicating that the rescuability boundary—the confidence level below which pure-wrong examples are likely to be corrected by attention—is architecture-invariant. Disruption thresholds span

Table 22: Within-architecture prediction of attention effects on LAMBADA (1.3B). AUC derived analytically from Cohen's d.

Architecture	Rescue AUC	d	Disruption AUC	d
GDN (h12-1)	.701	0.75	.732	0.87
RetNet (h3-1)	.696	0.73	.722	0.83
HGRN2 (h3-1)	.708	0.77	.743	0.92
Mean	.702	—	.732	—

a 47% relative range (GDN: 0.70, RetNet: 1.00, HGRN2: 1.14), meaning that the same confidence level on a pure-correct example may predict vulnerability in one architecture but safety in another. This asymmetry constitutes a second dimension of multiplicity beyond per-example disagreement: all three architectures agree on which wrong answers are rescuable, but disagree on which correct answers are vulnerable to disruption.

Table 23: Gaussian-intersection prediction thresholds ($|\log p|$) across architectures. Relative spread: range/mean.

	GDN	RetNet	HGRN2	Range	Rel. spread
Rescue	3.32	3.29	3.31	0.03	1.0%
Disruption	0.70	1.00	1.14	0.44	46.7%

Q Calibration Analysis

Cross-architecture disruption threshold variation (§4.1) could in principle arise from calibration differences rather than genuine differences in per-example processing. We compute expected calibration error (ECE; Guo et al. 2017) for each architecture's h3-1 model on LAMBADA using 15 equal-width bins. ECE values are similar: GDN 0.222, RetNet 0.224, HGRN2 0.235. The cross-architecture ECE spread (17%, computed as range/mean) is less than half the disruption threshold spread (47%; Table 23), indicating that calibration differences are insufficient to account for the observed threshold divergence. Rescue thresholds, which show only 1% cross-architecture spread, are consistent with near-identical calibration in the low-confidence regime.

R BoolQ Supplementary Analysis

BoolQ ($N = 3{,}270$; comprehension, binary yes/no classification) shows the highest ambiguity among the benchmarks we analyzed ($\alpha \approx 0.45$), but the Jaccard significance pattern is mixed: only 12 of 24 architecture $\times$ attention-ratio pair combinations yield $p < .01$. In particular, the RetNet–HGRN2 pair produces 0/8 significant Jaccard values, with several normalized effect sizes below zero, suggesting anti-correlated per-example effects for this pair.

Binary classification baseline. BoolQ is binary classification (two answer choices), whereas LAMBADA, HellaSwag, and ARC-Easy involve open vocabulary prediction or four-way multiple choice. Under a binary task, the null Jaccard baseline J_{null} is higher because chance overlap between helped/hurt sets is larger when there are fewer ways to be wrong. This elevates the significance threshold: observed Jaccard must be substantially higher to exceed the null, which explains why 12/24 pairs fail to reach significance despite $\bar{J}_h = 0.103$ and $\bar{J}_u = 0.098$—values comparable to the four main benchmarks.

Per-example summary. Table 24 reports the key metrics. The high α reflects that BoolQ examples are more sensitive to architectural variation in the presence of attention: approximately half of all examples change correctness status for at least one architecture. However, the mixed Jaccard significance prevents us from confidently concluding that the cross-architecture overlap structure matches the pattern observed on the four main benchmarks.

Table 24: BoolQ predictive multiplicity at 1.3B (h3-1).

	N	α	$\bar{J}_h$	$\bar{J}_u$	Sig. pairs	ρ_α
BoolQ	3270	.45	.103	.098	12/24	+0.1

RetNet–HGRN2: 0/8 significant; multiple normalized ES < 0.

The significant Jaccard overlap concentrates in GDN–RetNet pairs (8/8 significant), while RetNet–HGRN2 shows none (0/8). This pair-specificity does not appear on the four main benchmarks, where all architecture pairs show significant overlap; it may reflect a BoolQ-specific interaction between binary classification structure and particular recurrent architectures rather than a general property of architecture pairs.

Confidence margin. Confidence margin analysis on BoolQ shows the rescue pattern (C2) partially holds (2/3 architectures significant, $d = 0.16$–0.36) but the disruption pattern (C1) is absent (0/3, $|d| \leq 0.10$), consistent with compressed log-likelihood ranges in binary classification reducing

the predictive power of confidence margins. This extends the gradient observed across LAMBADA, ARC-Easy, and HellaSwag: C1 effect sizes decrease from open-ended prediction ($d \approx 0.9$, LAMBADA) through ARC-Easy ($d \approx 0.1$–0.3) to HellaSwag ($d \approx 0$ under `acc_norm`) and binary classification ($d \approx 0$, BoolQ). The gradient broadly tracks the coefficient of variation (CV) of the log-likelihood confidence proxy: LAMBADA (CV $=$ 0.75) $>$ ARC-Easy (0.54) $>$ HellaSwag (0.49) $>$ BoolQ (0.22). BoolQ's high-frequency short completions ("yes"/"no") yield elevated baseline log-likelihoods (mean $|\text{loglik}| = 3.04$ vs. 0.84 for LAMBADA) with similar absolute spread (IQR ≈ 0.9), compressing the relative variation and rendering the confidence margin unable to separate borderline-correct from robustly-correct examples.

Interpretation. The RetNet–HGRN2 anti-correlation may reflect different error modes specific to binary classification: when one architecture's recurrent state resolves a yes/no ambiguity through attention, the other architecture's resolution may be disrupted by the same attention intervention. This pattern warrants further investigation with additional binary classification benchmarks to determine whether it is BoolQ-specific or characteristic of two-choice tasks more broadly.

S WinoGrande Supplementary Analysis

WinoGrande ($N = 1{,}267$; reasoning, two-choice coreference resolution) shows high ambiguity ($\alpha \approx 0.41$ at h3-1) but mixed Jaccard significance: 10 of 24 architecture $\times$ attention-ratio pair combinations yield $p < .01$. Mean pairwise Jaccard ($\bar{J}_h = 0.110$, $\bar{J}_u = 0.103$) is comparable to the four main benchmarks, but the elevated null baseline from near-chance accuracy ($\sim$60%) reduces statistical power to detect above-chance overlap.

Near-chance accuracy. WinoGrande accuracy across hybrid configurations ranges from 59% to 62%, close to the two-choice chance level of 50%. At near-chance accuracy, a large fraction of examples are borderline—small perturbations (such as adding attention) can flip predictions in either direction, inflating the helped and hurt sets and consequently α. This structural property, shared with BoolQ ($\sim$59% accuracy, $\alpha \approx 0.45$), may partially explain the elevated α rather than task-intrinsic multiplicity. PIQA (73% accuracy, $\alpha = 0.14$) and HellaSwag (60% normalized accuracy, $\alpha = 0.15$) demonstrate that near-chance accuracy is not the sole driver, but we cannot fully exclude it as a contributing factor for WinoGrande and BoolQ.

Table 25: WinoGrande predictive multiplicity at 1.3B (h3-1).

	N	α	$\bar{J}_h$	$\bar{J}_u$	Sig. pairs	ρ_α
WinoGrande	1267	.41	.110	.103	10/24	-0.9

α variation across ratios: 0.054 (> 0.03 threshold of main benchmarks).

T 340M Per-Example Analysis

T.1 LAMBADA

At 340M, LAMBADA per-example multiplicity closely matches the 1.3B pattern: $\alpha = 0.313$ (vs. 0.307 at 1.3B), with mean pairwise Jaccard $\bar{J} \approx 0.089$ (vs. 0.095). Per-example directionality is preserved: HGRN2 shows net positive effects at h3-1 ($+238$ examples), while GDN (-256) and RetNet (-106) show net negative effects, mirroring the 1.3B profile (HGRN2 $+153$, GDN -153, RetNet -69).

However, aggregate convergence is absent at 340M: $\rho = -0.50$ ($p = .45$), with non-monotonic σ across attention ratios ($0.025 \to 0.012 \to 0.018 \to 0.011 \to 0.016$). This contrasts sharply with 1.3B ($\rho = -1.0$, $p = .017$, monotonic σ decrease). The decoupling between per-example divergence and aggregate convergence thus manifests differently across scales: at 340M, per-example multiplicity is present but aggregate convergence is not, suggesting that convergence requires sufficient model capacity while multiplicity does not.

Within-benchmark α variation across attention ratios is also larger at 340M (0.047, from 0.313 at pure to 0.266 at h3-1) than at 1.3B (< 0.03; §4.1), suggesting that smaller models are more sensitive to the attention ratio in terms of per-example disagreement patterns.

Table 26: 340M vs. 1.3B LAMBADA comparison.

	340M	1.3B
α (pure)	.313	.307
$\bar{J}$ (h3-1)	.089	.095
ρ_σ	-0.50	-1.0
p	.45	.017
GDN net	-256	-153
HGRN2 net	$+238$	$+153$

T.2 Cross-Benchmark Scale Comparison

Table 27 extends the scale comparison to all four main benchmarks. Two patterns emerge. First, α is systematically higher at 340M across all benchmarks, with the largest gap on ARC-Easy (+8.4pp) and the smallest on LAMBADA (+0.6pp). Second, pairwise Jaccard remains stable across scales ($\bar{J}_h = 0.08$–0.10 at both scales), confirming that the qualitative pattern of per-example multiplicity—which architectures affect which examples—is scale-robust even as its magnitude increases at smaller scale. Notably, the α inflation does not track accuracy loss: HellaSwag has the largest accuracy gap between scales yet the smallest $\Delta\alpha$ (+1.9pp), while ARC-Easy has a smaller accuracy gap but the largest $\Delta\alpha$ (+8.4pp), arguing against accuracy as the primary driver of scale-dependent multiplicity.

Table 27: 340M vs. 1.3B per-example multiplicity across four benchmarks. α: ambiguity at pure-recurrent configuration. Acc: mean accuracy across three architectures at pure-recurrent. $\bar{J}_h/\bar{J}_u$: mean pairwise Jaccard of helped/hurt sets. $\Delta\alpha$: 340M − 1.3B.

	340M				1.3B			
Benchmark	Acc	α	$\bar{J}_h$	$\bar{J}_u$	α	$\bar{J}_h$	$\bar{J}_u$	$\Delta\alpha$
PIQA	.663	.181	.090	.085	.143	.091	.094	+.038
HellaSwag†	.420	.167	.084	.083	.148	.095	.085	+.019
ARC-Easy	.594	.289	.098	.103	.205	.092	.099	+.084
LAMBADA	—	.313	.089	.090	.307	.095	.093	+.006

†`acc_norm`. LAMBADA accuracy omitted (evaluated in separate run).

Convergence is absent at 340M across all benchmarks (all $|\rho| \leq 0.4$, $p \geq .45$), in contrast to 1.3B where LAMBADA shows monotonic convergence. The absence of convergence at 340M extends beyond LAMBADA to HellaSwag ($\rho = 0.4$, $p = .52$), PIQA ($\rho = 0.3$, $p = .68$), and ARC-Easy ($\rho = -0.1$, $p = .95$).

U Subsample Sensitivity of α

To assess the reliability of α estimation at different sample sizes, we repeatedly subsample each benchmark (1,000 bootstrap replicates, seed 42) and compute the 95% CI width of α.

V Complementarity Decomposition

We decompose the three-architecture vote on each example into four categories: unanimous correct (3/3), majority correct (2/3), minority correct (1/3), and unanimous wrong (0/3). This decomposition explains the structural failure of majority vote despite substantial per-example complementarity.

Table 28: α estimation stability: 95% CI width by subsample size.

Benchmark	$n = 100$	$n = 200$	$n = 500$	$n = 1000$	Full N
PIQA	.130	.095	.066	.045	.034
HellaSwag	.140	.100	.062	.044	.013
ARC-Easy	.160	.105	.072	.050	.032
LAMBADA	.180	.125	.082	.057	.025

At $n = 100$, CI width (0.13–0.18) spans the entire four-benchmark α range (0.14–0.30), rendering α estimation unreliable. This justifies excluding COPA ($n = 100$) from the per-example analysis.

Table 29 reports the decomposition at h3-1 (25% attention). Majority vote benefits only from 2/3 examples (where the correct answer holds a majority) but is actively harmed by 1/3 examples (where the correct architecture is outvoted). On LAMBADA, HellaSwag, and ARC-Easy, the MV exploitation ratio—the fraction of complementary examples where MV selects correctly—hovers near 51%, indicating a near-even split between MV-exploitable and MV-hostile disagreements. The oracle, by contrast, exploits both categories, explaining the large oracle–MV gap (+7–14 pp).

Table 29: Complementarity decomposition at 1.3B h3-1. Fractions of total examples in each vote category. MV expl.: fraction of complementary examples where majority vote is correct.

Benchmark	3/3	2/3	1/3	0/3	MV expl.
LAMBADA	.361	.151	.145	.343	51.1%
HellaSwag	.519	.078	.073	.331	51.6%
ARC-Easy	.603	.107	.103	.187	50.9%
BoolQ	.376	.257	.197	.170	56.6%

This near-50% exploitation ratio is structurally invariant across attention ratios: on LAMBADA, the ratio ranges from 51.1% to 54.4% across all five configurations; on HellaSwag, from 50.3% to 51.6%. Attention changes neither the amount nor the structure of cross-architecture complementarity.

W Seed-Variation Calibration

To contextualize the 3× within- vs. cross-architecture Jaccard gap (§4.2), we compare against within-seed variation using the Pythia 70M (Biderman et al., 2023) PolyPythias models (van der Wal et al., 2025), which provide 9 individually trained seeds of the same architecture.

On LAMBADA ($N = 5{,}153$), within-seed mean Jaccard is $\bar{J}_h = 0.273$ (helped) and $\bar{J}_u = 0.393$ (hurt), compared to our cross-architecture reference $J \approx 0.10$. On ARC-Easy ($N = 2{,}376$), within-

seed $\bar{J}_h = 0.312$ and $\bar{J}_u = 0.354$, versus cross-architecture $J \approx 0.092$. Within-seed ambiguity is $\alpha = 0.36$ (LAMBADA) and $\alpha = 0.49$ (ARC-Easy), compared to cross-architecture $\alpha = 0.30$ and 0.21 respectively.

Seeds thus produce comparable or higher amounts of per-example disagreement (α) but on more overlapping examples (higher J). Cross-architecture variation produces per-example effects on less overlapping examples (lower J), consistent with architecture-driven divergence producing qualitatively different per-example patterns than seed noise. The within-seed Jaccard is 2.7–4.3$\times$ higher than cross-architecture Jaccard across both benchmarks.

Three caveats qualify this comparison. First, scale mismatch: Pythia 70M is $5\times$ smaller than our smallest models (340M); attempts to use Pythia 410M failed repeatedly. Second, architecture family: Pythia is a standard Transformer, not a hybrid recurrent architecture, so this is a cross-family reference point rather than a matched control. Third, accuracy regime: Pythia 70M accuracy (LAMBADA 0.17–0.23, ARC-Easy $\sim$0.38) is far below our models ($\sim$0.50, $\sim$0.72), and Jaccard is affected by base rates; the 2.7–4.3$\times$ ratio may differ at accuracy regimes closer to our main experiments. These limitations restrict the comparison to a reference point for seed-variation magnitude rather than definitive evidence for architecture-driven effects.

X Extended Architecture Validation (Five Architectures)

To test whether the low cross-architecture Jaccard pattern generalizes beyond the three primary architectures, we extend the analysis to include two additional architectures from the same model zoo: GLA (Gated Linear Attention) and DeltaNet, yielding ten pairwise comparisons. Both are available only at 340M; all results below use the h3-1 (25% attention) configuration.

Table 30: Pairwise Jaccard similarity across five architectures (340M, h3-1). All 40 permutation tests yield $p \leq .002$ ($z = 3.49$–12.21).

Benchmark	Pair	J_h	J_u
LAMBADA	GDN–RetNet	.128	.121
	GDN–HGRN2	.123	.083
	GDN–GLA	.124	.100
	GDN–DeltaNet	.115	.100
	RetNet–HGRN2	.108	.083
	RetNet–GLA	.123	.107
	RetNet–DeltaNet	.115	.096
	HGRN2–GLA	.127	.086
	HGRN2–DeltaNet	.125	.090
	GLA–DeltaNet	.134	.081
	Mean	*.122*	*.095*
ARC-Easy	GDN–RetNet	.089	.114
	GDN–HGRN2	.122	.105
	GDN–GLA	.114	.106
	GDN–DeltaNet	.084	.102
	RetNet–HGRN2	.087	.115
	RetNet–GLA	.112	.110
	RetNet–DeltaNet	.138	.123
	HGRN2–GLA	.098	.087
	HGRN2–DeltaNet	.109	.088
	GLA–DeltaNet	.129	.098
	Mean	*.108*	*.105*

All pairwise Jaccard values remain ≤ 0.14, consistent with the three-architecture results at 1.3B (Table 9). The seven new pairs involving GLA and DeltaNet show no systematic difference from the original three pairs (Mann-Whitney $p > .3$ for both helped and hurt), indicating that the low-overlap pattern is not specific to the original architecture triple. A sixth architecture (HGRN; Qin et al., 2023) is also available in the zoo but excluded from the main analysis: its substantially lower baseline accuracy on LAMBADA (0.263 vs. 0.32–0.38 for the other five) mechanically inflates helped-set sizes and Jaccard values. All 15 six-architecture permutation tests nonetheless remain significant ($z \geq 2.9$, $p \leq .006$). The same pattern holds on HellaSwag ($N = 10{,}042$) and PIQA ($N = 1{,}838$): all 20 additional pairwise tests yield $p < .01$ with mean $J_{\text{helped}} \leq 0.10$. Full four-benchmark and six-architecture data are included in our code release.